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Bus maintenance scheduling using multi-agent systems

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Abstract

Real-world scheduling problems are usually complex and involve many approaches to find sub-optimal rather than optimal solutions using reasonable computing resources. The bus maintenance scheduling problem, which is distributed and dynamic in nature, has received less attention compared to scheduling problems in manufacturing. The characteristics of bus maintenance scheduling problems are first identified, then a multi-agent system (MAS) is proposed to heuristically solve the bus maintenance scheduling problem investigated here. The methods for which the model caters for distributed and dynamic problem solving are then discussed. The model is then tested and generates solutions with equal optimality to reported studies and requires less computing time without constraint violation, and is comparable to the work of a mathematical programming approach. Finally, the advantages of MASs are presented and further studies are identified.
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Keywords: Agent; Heuristic algorithm; Multi-agent system; Optimization; Scheduling

1. Introduction

In bus transportation companies, buses are maintained regularly in fixed time or mileage intervals in geographically distributed maintenance depots. A depot performs maintenance jobs occurring and arriving dynamically. The problem is dynamic in nature due to certain events: (1) the unexpected breakdowns of buses, (2) daily changes in the assignment of buses, (3) daily incoming maintenance tasks and (4) the change in availability of maintenance resources. Normally, a bus transit company schedules its incoming maintenance orders in advance to take place in a specific time period. Such a time period is regarded as the scheduling horizon and can be sub-divided into time slots. Each maintenance job can be scheduled into one or several time slots (Haghani and Shafahi, 2002). However, this predictive schedule is subject to change during execution due to unforeseen events. Hence, bus maintenance scheduling is comprised of two different parts: (1) devising an initial schedule based on all known information about orders and their constraints, which is referred to as predictive scheduling, and (2) adapting the schedule to dynamic events in real time, or the so-called reactive scheduling problem.

Ideally, managers of bus transit companies aim to maintain buses during their idle time to avoid extracting them from service (street time). In practice, this ideal situation is subject to constraints such as the availability of maintenance bays and buses. A maintenance bay is capable of handling only certain types of maintenance work; a bus is not available to be worked on for a particular maintenance type when in an alternative bay, receiving another form of maintenance, or when in service. Given a set of bus maintenance tasks, of interest is (1) finding good schedules for all jobs to minimize the total out-of-service time and maximize the utilization of maintenance resources, and (2) to simultaneously accommodate the four aforementioned dynamic events.

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Three integer-programming models and several heuristic solution methods are proposed by Haghani (2002) to solve the first part of the problem. A simulation model is built to mimic the real-world maintenance practice for 182 days and a branch-and-bound algorithm is used to find optimal solutions for each day for comparison purposes.

The mathematical model is run from scratch daily to incorporate new information of dynamic events from the previous day. Obviously, this approach ignores unforeseen events happening in the current day and postpones their computation till the following day. Costs or equivalently financial losses are incurred in situations where buses fail to arrive at scheduled bays, or are idly waiting for maintenance.

The purpose of this paper is to propose a multi-agent model, which can not only generate good schedules for a set of maintenance jobs within reasonable computing time, but can also have potential to react to unforeseen events dynamically while realizing a certain level of optimization.

There are few reported studies to solve bus maintenance scheduling problems in dynamic and distributed environments. Intelligent agent techniques have been successfully applied in many fields of manufacturing, including dynamic and distributed scheduling problems. Hence, what is new here is the application of agent-based technology in the form of multi-agent systems (MASs) to bus maintenance scheduling. A MAS model built on ZEUS (ZEUS) demonstrates this distributed dynamic scheduling technique. The working of the model and how it generates schedules for a set of orders is presented; however, the detailed mechanism concerning agent co-ordination and reaction to dynamic events will be the subject of further research.

In the following sections, the analysis of the problem is first presented and the MAS model for bus maintenance scheduling is described. The methods in which the model caters for distributed and dynamic problem solving are then discussed and the schedules generated by the MAS method are compared with those obtained using the mathematical programming approach of Haghani (2002). Finally, the advantages of the model and proposed future studies are presented.

2. Characteristics of the bus maintenance scheduling problem

Scheduling involves the allocation of resources to tasks over time subject to temporal and capacity constraints. It is a decision-making process with the goal of optimizing one or more objectives (Pinedo, 2002). Generally, it can be formulated as an optimization problem where each feasible solution has an associated value and the objective is to find the feasible solution with the most optimal value (Cormen et al., 2001). In many scheduling problems, the computing time grows exponentially with an increase in the number of jobs (buses to be maintained) and resources (bays and maintenance facilitates). Subsequently, in practice, researchers are more interested in finding sub-optimal solutions within reasonable computing time rather than in finding optimal solutions, which may take longer to generate but with only a limited increase in optimality. This paper explores another promising approach by using a MAS system to heuristically solve the bus maintenance scheduling problem. The notion of a bus maintenance schedule refers to the assignment of resources to a task in one or several connected discrete time intervals. An order requires one maintenance schedule and each bus may have many maintenance and service schedules which are not permitted to conflict with each other. The whole scheduling horizon may include many orders or equivalently maintenance schedules.

2.1. The problem details

The bus maintenance scheduling problem here concerns a depot with several bays, where each bay is responsible for certain maintenance types and each bus requires several maintenance jobs to be performed in bays within preset time intervals or windows: the time intervals in which maintenance should begin. The Bus-Status Tracking System gathers and sends orders to a depot, which then requests the service schedules of corresponding buses from the Bus-Service Planning System as illustrated in Fig. 1. Orders are to be scheduled over all bays in order to realize their maintenance objectives while meeting both temporal and capacity constraints.

The two specific objectives of the problem are: (1) to minimize the interruptions in the bus operating/service schedule, and (2) to maximize the utilization of the maintenance facilities. The time constraints include: (1) bus-service schedules—buses need to be on the street during certain intervals, (2) bus maintenance schedules—buses are required to be in certain bays at

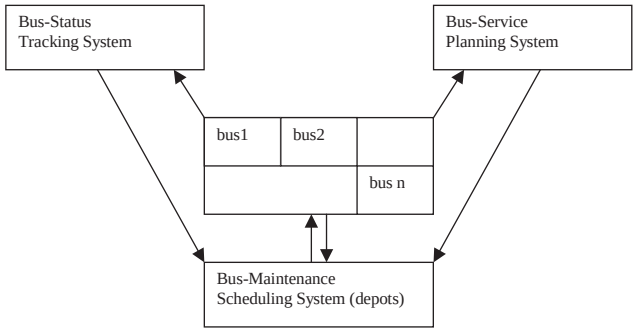


Fig. 1. The systems involved in bus maintenance scheduling.

specific times and (3) bus availability—buses cannot be in more than one bay simultaneously. The resource constraint concerns bay availability—only one maintenance job can be performed in a bay at a certain time. Bus-service schedules are the only constraints that can be relaxed, the "disturbance time" of which will be minimized as one of the objectives of the model. Here "disturbance time" refers to the time period during which a bus is to be extracted from service in order to meet its maintenance requirement. Thus, a spare bus should be arranged in place of that extracted to complete/continue the service. Hence, the first part of the bus maintenance problem may be considered as a one-stage scheduling problem with parallel machines and uncertain release times and its distinctive characteristics involve dynamically incoming orders and distributed working locations.

Assumptions: The modeling and computational assumptions are as follows:

- Each maintenance job is an entity, which can be fully completed in one bay.
- There is no preemption, each job must be completed once it starts.
- No cancellation is permitted.
- No two different types of jobs can be carried out simultaneously in the same bay.

Performance measurements: The performance of the job scheduling activity is measured based on two characteristics:

- The quality of its solution—the total violation of bus service time.
- The computing time required to obtain the solution.

3. The MAS model for the bus maintenance scheduling problem

An agent can be regarded as a self-directed software object with its own value system and a means to communicate with other agents (Baker, 1998). A MAS is defined as "a loosely coupled network of problem solvers that work together to solve problems that are beyond the individual capabilities or knowledge of each problem solver" (Durfee, 1988). The proposed model is a hierarchical MAS, with three layers and four types of agents to cooperate together to iteratively schedule incoming maintenance jobs. The initial schedules generated by this model could be considered as a starting point from which reactions to dynamic events could be based.

3.1. The agents in the model

Typically, an agent according to Jennings et al. (1998) has three features: (1) it can proactively realize goals, (2) it can perceive the changes to its environment as well as change its environment, and (3) it can communicate with other agents to realize its goals. There are four types of agents in the current model: (1) the Order Receiver agent, (2) Mediator agents, (3) Bay agents and (4) the Database agent (DA). Their respective roles are depicted in Fig. 2 and described below.

- *Order Receiver agent*: The Order Receiver agent is the only mechanism through which the model is connected to the outside world. It receives maintenance orders from the Bus-Status Tracking System and their related service schedule information from the Bus-Service Planning System. An order has information concerning bus identity, type of maintenance and the starting window. If a set of orders and their bus service schedules are all known before the scheduling work begins, they are then sorted based on their urgency or priorities by the Order Receiver agent and are passed to their respective Mediator agents. The Order Receiver agent releases the next order for scheduling after it receives the confirmed maintenance schedule for the previous one. It also stores the maintenance schedules of all scheduled orders.
- *Mediator agents*: One Mediator agent is responsible for only one type of maintenance order. Each Mediator agent keeps a name list of Bay agents, which are capable of the maintenance type that the Mediator agent manages. When an order from the Order Receiver agent arrives, the Mediator agent passes the order to the Bay agents on its name list and also to the DA. All bids from Bay agents are sent back to the Mediator agent, which then evaluates the bids and awards the order to the appropriate Bay agent and, at the same time, notifies the Order Receiver agent of the result; Mediator agents do not store any inputs or results.
- *Bay agent*: Bay agents represent the real maintenance bays with the goal to bid for as many orders as possible; they store maintenance jobs and work according to maintenance schedules. When a Bay agent is informed of a maintenance order for a particular bus, it will also be provided with all the existing maintenance schedules associated with this bus in other bays by the DA. The Bay agent can then

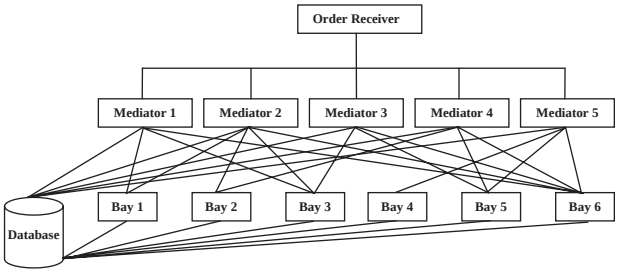


Fig. 2. The structure of the MAS.

determine a bid for the order intended for its Mediator agent given all the above constraints and release time windows, which are the time intervals in which a maintenance task should be completed for a certain bus. If the Mediator agent accepts the bid, it will send the Bay agent a confirmation message with information of a confirmed maintenance schedule. Then the Bay agent will store this schedule into its local Resource Database and notify the DA about the schedule. Otherwise, if the Mediator agent denies the bid, the Bay agent just ignores the order and deletes the proposed bid from its local database. Each Bay agent tries to maximize its utilization by proposing bids as early as economically possible. The bay-oriented complete schedules for the scheduling horizon can be generated using maintenance schedule records stored in the Bay agents.

- *Database agent*: The DA stores the maintenance schedules of each bus, which is the information needed by Bay agents for biding purposes. Once a Mediator agent receives an order for a bus, it will instruct the DA to search for all records of maintenance schedules associated with that bus. The DA then distributes these records to those Bay agents intending to bid for the order. The DA updates its records by receiving newly confirmed maintenance schedules from Bay agents.

3.2. The structure of the MAS

The MAS for bus maintenance scheduling is a hierarchical distributed agent-based system as shown in Fig. 2, which receives dynamic information as inputs from the Bus-Service Planning System and the Bus-Status Tracking System (Fig. 1). The system iteratively produces maintenance schedules for all buses with collaboration amongst the agents. The MAS has no centralized computing unit, instead, computing tasks are decomposed and distributed to the four agent types with features such as bidding, evaluating, schedule storing and database updating. Bay agents make bids based on their local knowledge and the best bids are chosen by Mediator agents with a global view.

3.3. Scheduling process

The Order Receiver agent passes order $O_{ijk} = (j, mt_1, mt_2, m_k)$, $i \in [1,..,I]$, $j \in [1,..,J]$, $k \in [1,..,K]$ of bus j for maintenance type m_k with starting and ending maintenance times mt_1 and mt_2, respectively, to mediator M_k, which then forwards it to several Bay agents that are capable of doing the job and invites them for bids. Here, I, J, K, represent the totals of orders, buses and maintenance types, respectively. Mediator M_k simultaneously instructs the DA to supply all the maintenance schedule records associated with bus j to these Bay agents. Each Bay agent determines its bid locally based on the following information: (1) its capability, that is, how many time slots t, are already occupied, (2) the maintenance schedules MS_{jl} of bus j in bay l, of a total of L bays where $l \in [1,..,L]$, (3) the service schedules $\mathrm{SS}_j = (j, st_1, st_2)$ of bus j with starting and ending times st_1 and st_2, respectively, and (4) the maintenance starting time windows mt_1 of bus j. The bids are the earliest time slots that can be allocated to the bays and are sent back to mediator M_k, which evaluates the bids, changes the best bid to a confirmed maintenance schedule and sends the confirmed schedule to both the winning Bay agent and the Order Receiver agent. The winning/successful Bay agent stores the confirmed schedule in its local database and notifies the DA to record the new schedule. The losing Bay agents receive denied messages from mediator M_k and ignore the order. The system progresses to the next order when the Order Receiver agent receives the confirmed schedule. The order distribution heuristic flow chart is as shown in Fig. 3 and indicates the relationship among the four types of agents: Order Receiver, Mediator, Bay agents and DAs.

4. Model implementation and methodologies

The multi-agent bus maintenance scheduling system is built on the ZEUS (ZEUS) platform using JAVA (JAVA) and the communication among agents is based on the Contract Net Protocol (CNP) (Smith, 1980). The basic components created automatically by ZEUS for an agent include the Mailbox and Mail Handler, the Planner, the Co-ordinator and the local Databases. In the present model, agents do not use the Planner and Co-ordinator frequently since their decision making is directed through external programs. The workings of the agent implementations are described below:

- Order Receiver agent: The Order Receiver agent sorts orders according to their priority. The most urgent order is served with the highest priority. The urgency of order O_{ijk} in a list is defined as the ratio of the maintenance time span and the processing time required, that is

$$U_i = \frac{D_i - S_i}{P_i}, \tag{1}$$

where U_i is the urgency of order O_{ijk}, D_i is the maintenance span starting window, S_i includes one or several service windows which overlap the starting window and P_i is the processing time required.
The difference $D_i - S_i$ is the slack time of order O_{ijk} and represents how many time slots might be available for allocation to the order: the larger the

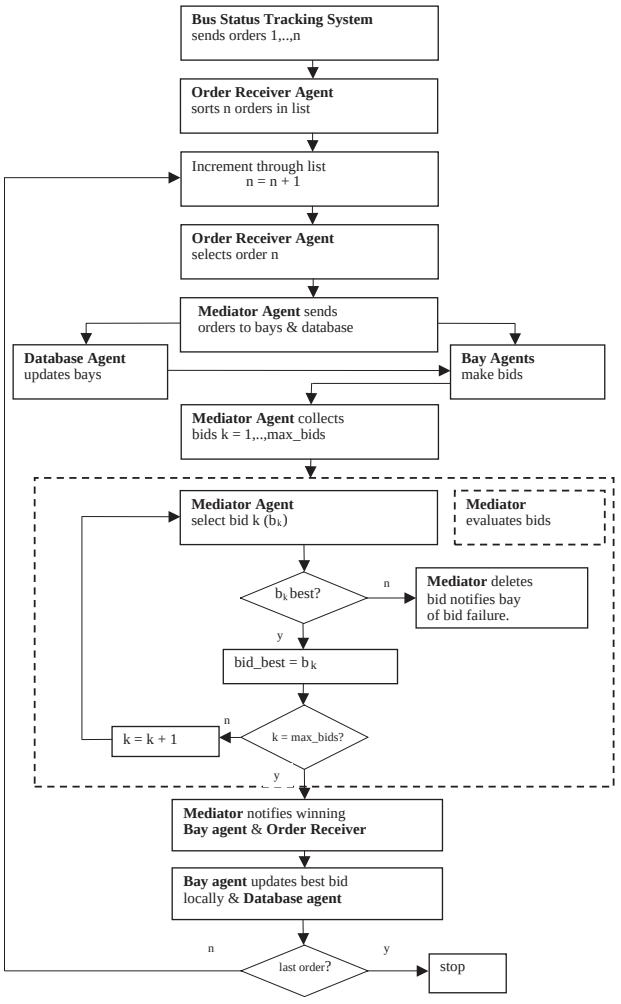


Fig. 3. Order distribution heuristic flow chart indicating the relationship of the four types of agents: Order Receiver, Mediator, Bay agents and DAs.

slack time, the greater the chance the order is allocated to the available time slots. However, if order O_{ijk} has a long processing time P_i, it may be more difficult to allocate maintenance resources or equivalently maintenance bays. Hence, U_i may be interpreted as the relative slack time of order O_{ijk} and its value reflects the urgency of an order. Subsequently, a smaller U_i implies a greater urgency and should be scheduled with higher priority. The algorithm for the Order Receiver agent to sort orders is shown in Fig. 4.

- Mediator agents: The computing work of a Mediator agent is to evaluate the bids and choose the one that causes minimal disturbance time and starts as early as possible. A bid includes information concerning (1) the minimal disturbance-time t_{dj}, which denotes the length of time bus j is extracted from service by the current maintenance schedule MS_{jl} and (2) the earliest starting time e_j of the maintenance schedule. A Mediator agent chooses bids with the smallest disturbance-time t_{dj}, and if bids have equal values of t_{dj}, the one with the earliest starting time e_j is chosen. If bids are equal in both t_{dj} and e_j, a Mediator agent will use its knowledge about the priorities of bays to choose a winner. For example, Bay 6 is able to cater for all five types of maintenance while Bay 4 can only cater for one type, as shown in Fig. 5. Subsequently, orders may only be offered to Bay 6 when no other bays can offer better solutions, while Bay 4 can win an order once it is able to provide a schedule of equal quality as any other favourable/best bids. Thus, Bay 6 has lower priority to win an order while Bay 4 has higher priority. This strategy can reduce congestion of a particular bay. The above procedure is summarized in pseudo code as shown in Fig. 6.
- Bay agent: The essential scheduling work is distributed to all bay agents whose aim is to find the earliest starting times with minimum service disruption. The algorithm used by Bay agents to find the best time periods for an order is shown in Fig. 7.
- DA: The computing work of the DA is to store all confirmed maintenance schedules associated with each bus and retrieve these records when instructed by Mediator agents. The bus-oriented schedules can be generated from information stored in this DA.

```
for all orders
        read bus maintenance order
        evaluate order urgency u[i]
end
quicksort orders in ascending order
```

Fig. 4. Order Receiver agent pseudo-code indicating order sorting.

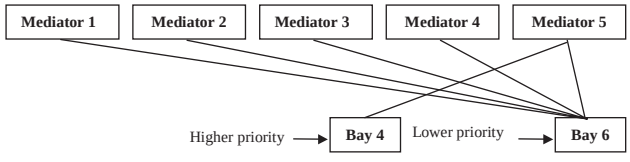


Fig. 5. The relationship of the capability and the priority of a bay.

```
for all bids
  read bid
  select bid with min disturbance time & earliest start
time
  if(disturbance_time[i] == min_disturbance_time)
    if(start_time_bid[i] < min_start_time)
      bid_best = bid[i]
    else if(start_time_bid[i] == min_start_time)
      use mediator agent knowledge to select best_bid
        endif
  endif
 end
```

Fig. 6. Mediator agent pseudo-code indicating bid selection.

```
for all bus maintenance schedules
    read schedule from database agent
    check whether bus service needs to be disrupted
    if (no service disruption)
       find earliest maintenance start time
    else
       find earliest maintenance start time to min disruption
    endif
end
```

Fig. 7. Bay agent pseudo-code to find best bid using local knowledge.

4.1. Agents co-operate to find global optimal schedules

To minimize the duration of being extracted from service, a four-part strategy is employed and involves: (1) the dispatching of jobs according to their urgency, (2) the scheduling of jobs in their idle times, (3) the scheduling of jobs as early as possible within their given maintenance time windows, and (4) the selection of the earliest bid, with minimal service disturbance time.

To maximize the utilization of maintenance resources (bays), orders are scheduled in adjacent time slots to save space in bays for future orders. More criteria may be added into decision-making agents (Mediator agents and Bay agents) as more domain knowledge is obtained.

Table 1
Comparing the maintenance starting times of each maintenance order in the two approaches

Bus no.	Type 1	Type 2	Type 3	Type 4	Type 5
(*a*) *The maintenance starting time of each order for the MAS approach*					
1	(5)	23	26	1	68
2	33	15	50	29	18
3	7	20	5	27	45
4	52	46	35	1	54
5	51	35	4	23	46
6	49	39	20	26	42
7	70	2	24	20	(50)
8	47	23	15	1	(26)
9	2	20	9	5	24
10	26	5	50	30	8
(*b*) *The maintenance starting time of each order* (*Haghani, 2002*)					
1	(20)	25	23	2	68
2	33	19	50	68	23
3	(10)	23	20	6	29
4	68	56	35	1	46
5	46	35	4	25	49
6	56	50	22	41	45
7	70	20*	26	22*	72
8	50	26	17	1	20
9	5	20	24	7	71
10	30	3	52	7	44

5. Model testing

Data from Haghani (2002) are used in order to compare the results of the Integer programming approach and the MAS approach pursued here. Orders are sequentially passed in the scheduling process according to Fig. 3, that is, one order must be completed before the next one starts to avoid retrieving inconsistent records of a certain bus from the DA.

There are in total 50 orders for 10 buses each with five types of maintenance. Six bays are under the co-ordination of five Mediator agents. The capabilities of each bay given in Table 3 of Haghani (2002) are built in the network of the MAS model (Fig. 2). Service windows given in Table 1 of Haghani (2002) for 10 buses and processing times of different types of orders given in Table 4 of Haghani (2002) are known in advance and are passed to all Bay agents by the Order Receiver agent. The release time slots given in Table 2 of Haghani (2002) are included in the orders. The testing scheduling horizon is 72 h, 1 h for each time slot, and the results of scheduling 50 orders are displayed in Tables 1 and 2.

The maintenance starting times of all 50 orders are listed in Table 1. For example, the maintenance order for bus 1 of type 1 will start in time period 5 in the MAS approach, but in time period 20 in the mathematical approach (Haghani, 2002). The total disturbance time of the service schedule can be evaluated from this table. The utilization and numbers of bays assigned to each maintenance type for each time slot are presented in Table 2.

5.1. Performance

- Solution quality: The schedule generated by the MAS violates a total of 3 h of service time, of which a 1-h violation from the order of bus 1 for maintenance type 1 is inevitable, since there is only a 1-h time interval for this maintenance task (see Tables 1 and 2 in Haghani (2002)). The same amount of total violation happens in the mathematical approach (Haghani, 2002). The violating starting times are indicated in Tables 1(a) and (b) with parentheses. The schedule by the MAS never violates the hard constraint—a bus cannot be present in two different bays, whereas the schedule in Haghani (2002) does violate it once, which is indicated in Table 1 (b) with an asterisk. The results show that the MAS approach can provide at least as good a solution as the mathematical programming approach of Haghani (2002).
- Computing time: The Order Receiver sorts orders with a worst case running time of complexity $O(n^2)$, where n is the order size. The Mediator agent requires a constant computing time and the Bay agents require

Table 2
The number of bays assigned to each maintenance type for each time period

Day 1—time of day inspection						Day 2—time of day inspection						Day 3—time of day inspection					
Hour	1	2	3	4	5	Hour	1	2	3	4	5	Hour	1	2	3	4	5
1				3		25		2	1	1	1	49	1				2
2	1	1		3		26	1		1	2	2	50	1		2		2
3	1	1		3		27	1		1	2	2	51	1		2		1
4		1	1	3		28				2	2	52	2				1
5	1	1	2	1		29				3	1	53	1				1
6	1	1	1	1		30				3	1	54					2
7	1	1		1		31				2		55					1
8	1			1	1	32				2		56					1
9			1		1	33	1			1		57					1
10			1		1	34	1					58					1
11					1	35		1	1			59					
12					1	36		1	1			60					
13						37		1				61					
14						38						62					
15		1	1			39		1				63					
16		1	1			40		1				64					
17		1				41		1				65					
18					1	42					1	66					
19					1	43					1	67					
20		2	1	1	1	44					1	68					1
21		2	1	1	1	45					2	69					1
22		2		1	1	46		1			3	70	1				1
23		2		2		47	1	1			2	71	1				1
24		2	1	1	1	48	1	1			2	72					1

a computational complexity of $O(n)$. Consequently, the MAS approach takes polynomial growing computing time, with an increase in the number of buses, excluding the time spent in communication among agents.

6. Advantages of the MAS approach

The main advantage of the MAS approach is that it can generate a schedule within reasonable computing time while maintaining a certain level of optimality. Such an advantage makes it a promising approach for dynamic, large size and distributed problems such as the bus maintenance scheduling problem.

6.1. At the predictive scheduling stage

All information for scheduling is known at this stage. The MAS guarantees a polynomial growing computing time as the number of buses increases while the computing time of mathematical approach grows exponentially.

Generally, the modelling of a mathematical approach may suffer from simplifying assumptions or insufficient consideration of constraints, which may lead to inapplicable solutions. In fact, the modifications to the aforementioned problems may result in a greater number of constraints, which may subsequently cause existing solution methods to be inefficient in practice. The MAS approach, however, does not require these assumptions since each agent makes decisions based on local knowledge of its environment and capability. The change of the constraints among agents will not affect the polynomial growing computing time of MAS since the computing tasks are distributed among many agents. For example, if the first assumption in Section 2.1 is relaxed, the mathematical model will change from a one-stage scheduling problem to a more complex job-shop problem. However, the changes in the MAS could include order dispatching rules adjusted in the Order Receiver agent, problem decomposition strategies in Mediator agents and the co-operation among Bay agents under the co-ordination of Mediator agents. The solution could still be guaranteed to be within polynomial growing time through the co-operation among agents.

The solutions of MAS approaches can be optimized at both the local and global levels. Firstly, a best local bid based on local information is proposed, then an agent with a global view of the system finds the best global solution from all those best local bids. The tests in the above sections demonstrate that such an approach can generate a quality predictive schedule.

6.2. At the reactive scheduling stage

A MAS is even more promising for solving reactive scheduling problems due to its ability to rapidly react to unforeseen events. Generally, approaches for reactive scheduling face two challenges: firstly the approach needs to revise existing schedules within a reasonable time to accommodate dynamic events, and secondly, it should do so respecting already made preparations for previously scheduled work whilst optimizing the schedule.

Some approaches accommodate for new events periodically, for example, scheduling procedures run daily from scratch to generate an optimized schedule for the next day. However, such practice might ignore already made preparations for previous scheduled work, which could result in instability and lack of continuity in schedules (Girsch et al.). Further, it might ignore the dynamic events occurring between two scheduling procedures and may cause a reduction in efficiency or loss of optimality when immediate attention is required. Other approaches involving dispatching rules, which are widely used in industry, may react to unforeseen events promptly but lack optimization of their schedules.

The MAS approach, however, has more flexibility in accommodating dynamic events on-line, while keeping a certain level of optimality due to its rapid reaction and co-ordination among agents, the detailed strategies of which will be explored in further research.

7. Conclusions and further studies

This paper presents a MAS approach to solve a bus maintenance scheduling problem, which is comprised of two sub-problems, predictive and reactive scheduling. The MAS is built to heuristically schedule maintenance jobs at the predictive scheduling stage and has a hierarchical structure that iteratively processes sorted maintenance orders. The methodologies to optimize schedules are implemented in each agent and in the co-operation among agents. An example was tested to show that the MAS could generate good solutions for the predicative scheduling problem. The complexity analysis shows that the whole schedule can be generated in an $O(n^2)$ worst time. Further, the MAS approach has the potential to solve dynamic scheduling problems.

Further studies will include extending the model to handle real-time dynamic events, which requires sophisticated co-ordination or negotiation among agents in order to reach local goals and globally optimal solutions. Mathematical programming could also be considered to optimize solutions at the local agent level. The extension may also include integrating the scheduling work in all maintenance depots in one bus transit company, or integrating a bus maintenance scheduling MAS with a bus service planning system. More constraints such as worker costs and tool availability and usage may need to be considered in real-world situations.

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